



BOARD QUESTION PAPER: FEBRUARY 2026

CHEMISTRY

Time: 3 Hrs.

Max. Marks: 70

General Instructions:

The question paper is divided into **four** sections.

- (1) **Section A:** Q. No. 1 contains **Ten** multiple choice type of questions carrying **One** mark each.
Q. No. 2 contains **Eight** very short answer type of questions carrying **One** mark each.
- (2) **Section B:** Q. No. 3 to Q. No. 14 are **Twelve** short answer type questions carrying **Two** marks each. (Attempt **any Eight**)
- (3) **Section C:** Q. No. 15 to Q. No. 26 are **Twelve** short answer type questions carrying **Three** marks each. (Attempt **any Eight**)
- (4) **Section D:** Q. No. 27 to Q. No. 31 are **Five** long answer type of questions carrying **Four** marks each. (Attempt **any Three**)
- (5) Use of log table is allowed. Use of calculator is not allowed.
- (6) Figures to the right indicate full marks.
- (7) For multiple choice type questions, only the first attempt will be considered for evaluation.
- (8) Physical constants:
 - i. Avogadro Number = $N_A = 6.022 \times 10^{23}$
 - ii. Gas constant = $R = 0.08206 \text{ L - atm. K}^{-1} \text{ mol}^{-1}$
 - iii. 1 Faraday = $F = 96500 \text{ Coulomb}$

SECTION – A

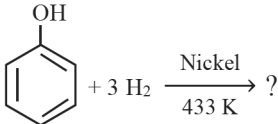
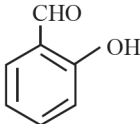
Q.1. Select and write the correct answer for the following multiple choice type of questions: [10]

- i. The electrolyte used in $\text{H}_2 - \text{O}_2$ fuel cell is _____.
(A) aqueous KCl (B) aqueous KOH
(C) aqueous HCl (D) aqueous KNO_3
- ii. The oxidation state of bromine in HOBrO_2 oxoacid is _____.
(A) +7 (B) +5 (C) +3 (D) +1
- iii. Acetonitrile may be prepared by heating the following reactants:
(A) Ethyl chloride with alcoholic KCN (B) Ethyl chloride with alcoholic AgCN
(C) Methyl chloride with alcoholic KCN (D) Methyl chloride with alcoholic AgCN
- iv. The carbonated water is an example of _____.
(A) solid in liquid solution (B) liquid in liquid solution
(C) gas in liquid solution (D) liquid in gas solution
- v. The unit of rate constant is per second, the order of reaction is -
(A) Zero (B) First (C) Second (D) Third
- vi. A system releases 10 kJ of heat and performs 15 kJ of work on the surrounding. Hence the change in internal energy is:
(A) + 5 kJ (B) - 5 kJ (C) + 25 kJ (D) - 25 kJ
- vii. The co-ordination complex ions $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]^{2+}$ and $[\text{Co}(\text{NH}_3)_5(\text{ONO})]^{2+}$ are _____ of each other.
(A) ionization isomers (B) solvate isomers
(C) linkage isomers (D) co-ordination isomers
- viii. The correct order for basic strength of amines and ammonia is _____.



- (A) $\text{NH}_3 > \text{R} - \text{NH}_2 > \text{R}_2\text{NH} > \text{R}_3\text{N}$ (B) $\text{NH}_3 > \text{R}_3\text{N} > \text{R}_2\text{NH} > \text{R} - \text{NH}_2$
 (C) $\text{NH}_3 < \text{R} - \text{NH}_2 > \text{R}_3\text{N} > \text{R}_2\text{NH}$ (D) $\text{NH}_3 < \text{R} - \text{NH}_2 < \text{R}_2\text{NH} > \text{R}_3\text{N}$
- ix. Amongst the following 3d-series elements, having highest value of first ionization enthalpy is _____.
 (A) Zn (B) Cu (C) Co (D) Sc
- x. The pOH of 0.01 M HCl solution is _____.
 (A) 1 (B) 2 (C) 11 (D) 12

Q.2. Answer the following questions:**[8]**

- Write the name of product obtained, when selenium is treated with magnesium metal.
- Write the SI unit of cell constant.
- Write the name of radioactive element in lanthanide series.
- Complete the following reaction:

- Write the IUPAC name of:

- Copper has fcc structure with edge length 495 pm. What is the radius of copper atom in pm?
- Write the chemical name of Teflon.
- Write the name of nanoparticle which acts as highly effective bacterial disinfectant.

SECTION – B**Attempt any EIGHT of the following questions:****[16]**

- Q.3.** Define:
 a. Isomorphism
 b. Unit Cell
- Q.4.** Explain the trends in the following properties of group 17 elements:
 i. Atomic radii
 ii. Electronegativity
- Q.5.** Explain Homoleptic and Heteroleptic complexes with example.
- Q.6.** a. Write the reaction of bromine water on glucose.
 b. Define – Enantiomers
- Q.7.** i. Write example of one dimensional nanostructure of size less than 100 nm.
 ii. Write the hydrolysis reaction of ethyl methyl ether.
- Q.8.** Calculate the osmotic pressure of solution containing 0.822 gm of sucrose in 300 mL of water at 298 K.
 [Given: Molar mass of sucrose 342 g/mol, $R = 0.08205 \text{ dm}^3\text{atm. K}^{-1} \cdot \text{mol}^{-1}$]
- Q.9.** Write the mathematical equation of first law of thermodynamics for following processes:



- a. Isochoric process
- b. Adiabatic process

Q.10. Write two similarities and two differences between lanthanoids and actinoids.

Q.11. What is the action of following reagents on chlorobenzene?

- i. conc. HNO_3
- ii. fuming H_2SO_4

Q.12. Explain Cannizzaro reaction with suitable example.

Q.13. A chemical reaction occurs in two steps:

- i. $\text{NO}_2\text{Cl}_{(\text{g})} \xrightarrow{\text{slow}} \text{NO}_{2(\text{g})} + \text{Cl}_{(\text{g})}$
- ii. $\text{NO}_2\text{Cl}_{(\text{g})} + \text{Cl}_{(\text{g})} \xrightarrow{\text{fast}} \text{NO}_{2(\text{g})} + \text{Cl}_{2(\text{g})}$
 - a. Write down the rate law.
 - b. Identify the reaction intermediate.

Q.14. The standard potential of electrode $\text{Cu}^{++} (0.02 \text{ M}) / \text{Cu}_{(\text{s})}$ is 0.337 volt. Calculate its potential in volt.

SECTION – C

Attempt any EIGHT of the following questions:

[24]

- Q.15. i.** Derive relationship between relative lowering of vapour pressure and molar mass of non volatile solute.
- ii.** Write statement of second law of thermodynamics.

- Q.16. i.** Convert, acetamide into ethyl amine.
- ii.** Explain amphoteric nature of water.

Q.17. Convert the following :

- i. Ethanol into sodium ethoxide.
- ii. Phenol into o-phenol sulfonic acid.
- iii. Bromomethane into methoxyethane.

- Q.18. i.** What is peptide linkage?
- ii.** Write general characteristics of interhalogen compounds.

- Q.19. i.** Write observed electronic configuration of chromium ($Z = 24$).
- ii.** Calculate magnetic moment of Ti^{3+} by using spin only formula. (Z of $\text{Ti} = 22$)
- iii.** Define Green chemistry.

- Q.20. i.** Write examples of coordination metal complexes in biology.
- ii.** Calculate the work done in oxidation of 2 moles of SO_2 at 298 K, if $\text{SO}_{2(\text{g})} + \frac{1}{2} \text{O}_{2(\text{g})} \rightarrow \text{SO}_{3(\text{g})}$
[Given: $R = 8.314 \text{ J/K/mol}$]

- Q.21. i.** Write the formula to measure % atom economy according to green chemistry.
- ii.** Convert:
 - a. Benzene into benzaldehyde
 - b. Cyclohexene into adipic acid

- Q.22. i.** Write a reaction for preparation of Nylon-6.
- ii.** The salt of Sc^{3+} ion is colourless and the salt of Mn^{3+} ion is coloured. Explain.
[Z of $\text{Sc} = 21$ and Z of $\text{Mn} = 25$]

- Q.23. i.** Write the postulates of Werner's theory of co-ordination complexes.



- ii. What is the action of ethane-1,2-diol on acetone?
- Q.24.** i. Write Tollen's reagent test for acetaldehyde.
ii. Write the structure of zwitter ion of sulfanilic acid. Write the reaction between benzene diazonium chloride and phenol in alkaline medium.
- Q.25.** i. Define:
Common ion effect.
ii. Write preparation of glucose from starch.
- Q.26.** i. Draw the structure of 2, 4-dinitrophenylhydrazone of acetaldehyde.
ii. The rate of $A + B \rightarrow P$ is $3.6 \times 10^{-2} \text{ mol/dm}^3/\text{s}$. When $[A] = 0.2 \text{ M}$ and $[B] = 0.1 \text{ M}$. Calculate the rate constant if reaction is first order in B and second order in A.

SECTION – D

Attempt any THREE of the following questions:

[12]

- Q.27.** i. Define:
a. Isotonic solution
b. Molecularity of reaction
ii. State and explain Hess's law of constant heat summation.
- Q.28.** i. Define:
a. Schottky defect
b. Ferromagnetism
ii. Write the name of catalyst used in preparation of HDP.
iii. Draw structure of isoprene unit of natural rubber.
- Q.29.** i. Calculate effective atomic number of Fe^{2+} in $[\text{Fe}(\text{CN})_6]^{4-}$
[Given: $(Z = 26)$]
ii. Draw neat and labelled diagram of lead accumulator.
iii. Write two advantages of hydrogen-oxygen fuel cell.
- Q.30.** i. Convert, ethanoic acid to ethanol.
ii. Explain the mechanism of alkaline hydrolysis of bromomethane.
- Q.31.** i. Acetic acid is 5% ionised in its decimolar solution. Calculate the dissociation constant of acetic acid.
ii. Write chemical formula of epsom salt.
Give two uses of H_2SO_4 .